PC MOCK
March, 2026
MATHEMATICS
(CORE) 2
$2\frac{1}{2}$ hours

2

Name: ....

Class: .....

PREMPEH COLLEGE

MOCK EXAMINATION

MARCH 2026 GENERAL MATHEMATICS/ MATHEMATICS (CORE) 2 $2\frac{1}{2}$ hours

[100 marks]

Answer **ten** *questions in all.* **All** *the questions in Section* A *and* **five** *questions from Section* B.

Each question, all necessary details of working, including rough work, must be shown with answer.

Give answers accurately and to appropriate degrees of accuracy where applicable.

Graph papers are provided for your use in the examination.

The use of non-programmable, silent and cordless calculator is allowed.

Core Mathematics
MOCK 2026
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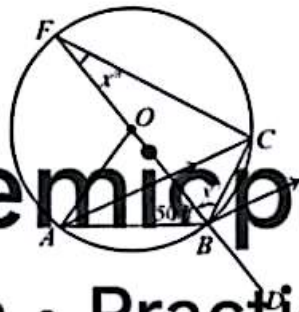
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SECTION A
[40 marks]

Answer **all** *the questions in this section.* **All** *questions carry* **equal** *marks.*

1. Two quantities x and y are such that y varies directly as the square of $(x + 1)$. Given that $y = 200$ when $x = 4$.
   a) find the equation connecting x and y
   b) determine the positive value of x when $y = 50$.

2. a) An amount of ₦12,500.00 is invested in a savings account which pays simple interest at a rate of 6% per annum. Calculate the balance accumulated by the end of 6 years.
   b) In the diagram, O is the centre of the circle. $\overline{BE}$ is a tangent to the circle at B. If $\angle ABF = 50^\circ$, $\angle CBF = y^\circ$, $\angle DFC = x^\circ$ and $AC//BE$, find the values of x and y.



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3. The base area of a cylinder opened at the top is 154 cm^2. If the height of the cylinder is 10 cm, find the:
   a) volume of the cylinder
   b) total area of the material used in making the cylinder.
   [Take $\pi = \frac{22}{7}$]

4. From a point on the ground, the angle of elevation of a drone is 45°. After the drone moves 20 metres horizontally away from the observer, the angle of elevation becomes 30°.
   a) Illustrate this information on the diagram
   b) Find, correct to **three** significant figures, the height of the drone.

5. a) Simplify $\frac{15x^2+xy-6y^2}{9x^2-4y^2}$
   b) A tourist has GH¢ 3600.00 for his expenses. If he extends his tour for 4 more days, he has to cut down his daily expenses by GH¢ 30.00, find the original duration of the tour.

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SECTION B

[60 marks]

Answer **five** *questions* **only** *from this section.* **All** *questions carry* **equal** *marks.*

6. a) Consider the following statements:

S_1: Poets are happy people

S_2: Every doctor is wealthy

S_3: No happy person is wealthy

(i) Draw a Venn diagram to illustrate the above statements.

(ii) Determine the validity of the following conclusions

(a) No poet is wealthy

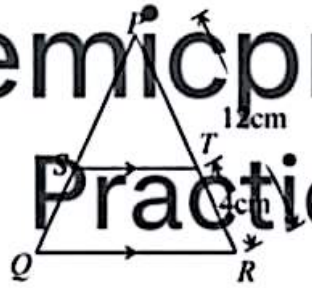
(θ) Doctors are happy people

b) A seven-sided polygon has three of its angles equal to θ and the others as $(\theta + 28)^0$, $3(\theta - 4)^0$, $(2\theta - 30)^0$ and $(134 - \theta)^0$. Find the value of θ.

7. a) Find the equation of a line passing through the points $A\left(1, -\frac{1}{2}\right)$ and $B(-5, 3)$.

b)

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The figure below shows a triangle PQR. $|PR| = 12\ cm$, $|TR| = 4\ cm$ and ST is parallel to QR. If the area of triangle PQR is 336 cm^2, find the area of the quadrilateral $QRTS$.

8. At a Music Concert, 60 participants had a choice of Country Music, Jazz or Rock Music. Out of these participants, 21 chose Country Music, 24 chose Jazz and 18 chose Rock Music. Also 3 chose Country Music only, 9 chose Jazz only, 2 chose Rock Music only and 5 chose all the three kinds of music.

a) Illustrate this information on the Venn diagram.

b) Use the Venn diagram to find the number of participants who chose

(i) Jazz and Country Music **only**;

(ii) exactly **two** kinds of Music;

(iii) **none** of the three kinds of Music.

9. a) Two locations, **A** and **B** on the same longitude are 3,940km apart. **A** is on Latitude 25^0N and **B** is on Latitude x^0S. Find, correct to the **nearest** degree, the value of x.
[Take $\pi = \frac{22}{7}$ and radius of the earth surface, $R = 6400\ km$]

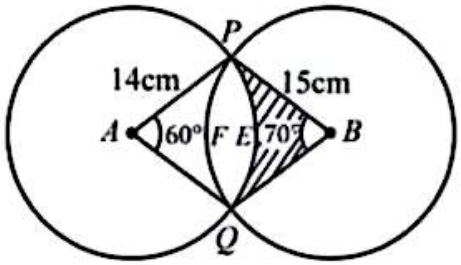
b) A clinic is on a bearing of 220^0 and 130^0 from a school and a market respectively. If the school is 70 metres away from the clinic and the market is 120 metres from the school, find, correct to

(i) **three** significant figures, the distance between the clinic and the market;

(ii) the **nearest** degree, the bearing of the school from the market.

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10. a) In the diagram below; $|AP| = 14\ cm$, $|PB| = 15\ cm$, $\angle PAQ = 60^0$ and $\angle PBQ = 70^0$. Find, correct to two decimal places, the area of the shaded region.
[Take $\pi = \frac{22}{7}$]



b) The sum of the squares of two consecutive positive even numbers is 244. Find the numbers.

11. a) The first three terms of an arithmetic progression (AP) are $8 - x$, $2x$ and $3x + 2$. Find the common difference of the AP.

b) A coin is tossed once and a six-sided die is also thrown.
(i) Draw a table showing all the possible outcomes.
(ii) Find the probability of obtaining;
(α) a number not more than 2 on the die;
(β) a head and a 5;
(y) a tail and a number not less than 3.

12. The table gives the marks obtained by a group of students in a test.

Marks	3	4	5	6	7	8
Frequency	5	$m-1$	m	9	4	1

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If the mean mark is 5,
a) calculate for the value of m
b) find the;
(i) mode;
(ii) median, of the distribution.
c) If one of the students is selected at random, find the probability that he scored at least 7 marks.

13. a) Without using tables or calculators, evaluate $\log\left(\frac{20}{9}\right) - \log\left(\frac{4}{125}\right) + 2\log\left(\frac{6}{5}\right)$

b) (i) Using scales of 2 cm to 2 units on both axes, draw two perpendicular axes Ox and Oy on a graph sheet for $-10 \leq x \leq 10$ and $-12 \leq y \leq 12$.
(ii) Draw on the same graph sheet the straight line joining the points A$(-3, -1)$ and C$(7, -1)$.
(iii) Construct the locus, L of all points equidistant from A and C.
(iv) Locate the point:
(α) B on L and above $\overline{AC}$ such that $|AB| = |BC| = 7.5\ cm$;
(β) D on L and below $\overline{AC}$ such that $|AD| = |CD| = 10\ cm$;
v) Join the points to form a quadrilateral ABCD. What is the name of the quadrilateral?

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